R script for data analysis of “Energy Prioritization and Neurometabolic Scaling in a Social Insect Brain”

Rstudio version: 2025.05.1

The following packages were used:

```
library(readxl)
library(RColorBrewer)
library(tidyr)
library(ggplot2)
library(ggpubr)
library(plyr)
library(dplyr)
library(MASS)
library(ucminf)
library(ordinal)
library(olsrr)
library(car)
library(agricolae)
library(lme4)
library(lmerTest)
library(performance)
library(FSA)
library(lmodel2)
library(smatr)
library(reshape2)
library(extrafont)
library(lsmeans)
```

```
require(flexplot)
library(stringr)
library(DHARMa)
library(AICcmodavg)
library(lattice)
library(mgcv)
library(ggpmisc)
library(survival)
library(survminer)
library(SurvRegCensCov)
library(MuMIn)
library(glmmTMB)
library(sjstats)
library(AER)
library(performance)
library(insight)
library(mixedpower)
library(data.table)
library(emmeans)
library(afex)
library(effectsize)
library(countES)
```

The following R code is grouped according to the data set used and question that is addressed. All data sets are initially read into R using the *read_excel()* function run within the *as.data.frame()* function. The name of the tab of the dataset excel file (submitted with the manuscript as "Dataset S1.xslx") and abbreviation used are included under each analysis for reference. For example, the R code for analysis of variation of worker mortality between treatments uses excel file tab "Mortality_data_figS11" and is abbreviated as "Mort" in the R code. Subtitles and notes for different components of each analysis are included with ## identifiers.

General code that isused for most analyses are included below separately from each analysis. Where dataframes, column variables or levels within columns are included in the code, these are written as "dataframe", "column" and "level", respectively. Examples are provided.

For each analysis, where the same code is used for analyzing different variables (e.g. for each environment, or for each treatment), an example of the code is provided for one of the variables. The variable of interest is **bolded** within the code and indicated in an associated note to facilitate reproducibility for the reader. To replicate the code for additional variables, simply replace the bolded variable with the desired alternative.

Subsetting Dataset by response variable levels

subset(dataframe, column==level)

ex. subset only laboratory colonies within the Mortality analysis dataframe

```
subset(Mort, Environment==”Lab”)
```

Obtaining counts per level in a column

```
dataframe%>%
group_by(factor(column)) %>%
tally()
```

ex. obtaining a tally of the number of workers for each environment in the Mortality dataframe

```
Mort%>%
group_by(factor(Environment)) %>%
tally()
```

Reorder levels within a column]

```
dataframe$column<-factor(dataframe$column, levels = c(“level 1”, “level 2”, “level 3”....))
```

ex. reorder environment levels from Lab,Field,Urban to Urban,Field,Lab in the Mortality dataframe

```
Mort$Environment<-factor(Mort$Environment, levels = c("Urban", "Field","Lab"))
```

Mortality Analysis

Dataframe: Mortality_data_figS11
Abbreviation for dataframe: Mort

```
##proportional hazards test for each treatment within laboratory colonies
MortLab<-subset(Mort,Environment=="Lab")
tstcox_surv <- coxph(Surv(day, status) ~ Treatment, data=MortLab)
cox.zph.fit<-cox.zph(tstcox_surv)
survdiff(Surv(day, status) ~ Treatment, data=MortLab)
ggcoxzph(cox.zph.fit)

##Weibull AFT regression for treatments pooled across environments
WeibullReg(Surv(day, status) ~ factor(Treatment)*Environment, data=Mort)
WeibullDiag(Surv(day, status) ~ factor(Treatment), data=Mort)

##Weibull AFT regression for treatments within laboratory colonies
WeibullReg(Surv(day, status) ~ factor(Treatment), data=MortLab)
WeibullDiag(Surv(day, status) ~ factor(Treatment), data=MortLab)

##Survival curves plotted by treatment and environment
mortplot1<-ggsurvplot(
  fit_surv1,
  data = Mort,
  linetype = c(1,1,1,5,5,5),
  size = .4,
  palette = c("cornflowerblue","darkblue","turquoise","cornflowerblue","darkblue","turquoise"),
  conf.int = TRUE,
  pval = F,
```

```
  break.time.by = 2,
  break.y.by = .1,
  legend=c("top"),
  xlab = "Days",
  #add_censor_mark(),
  ggtheme=theme(
    plot.margin = unit(c(1, 1, 1, 1), "cm"),
    panel.background = element_rect(fill="white",colour = "black", linewidth = 1, linetype = "solid"),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(size=15,colour = "black"),
    axis.title.x = element_text(size=20, vjust=-5),
    axis.text.y = element_text(size=15,colour = "black"),
    axis.title.y = element_text(size=20, vjust=5)))
mortplot1
```

Behavioral Response Assay

Dataframe: behavioral_response_figure_5
Abbreviation: Agg

```
##Calculate q1, median, q3 and IQR for each treatment
summary(Agg$ Behavioral_Response_Score)
tapply(Agg$Behavioral_Response_Score, Agg$Treatment, summary)
Aggmeds<-Agg%>%
  group_by(Treatment) %>%
  summarize(q1 = quantile(Behavioral_Response_Score,0.25),
            Med = median(Behavioral_Response_Score),
            q3 = quantile(Behavioral_Response_Score,0.75),
            IQR = IQR(Behavioral_Response_Score))
Aggmeds

##Levene's Test by environment
leveneTest(Behavioral_Response_Score ~ Environment, Agg)

##Kruskal Wallis Test by treatment
kruskal.test(max.score ~ Treatment, data = Agg)
pairwise.wilcox.test(Agg$ Behavioral_Response_Score, Agg$Treatment,
                     p.adjust.method = "BH")

#Box plot of behavioral response scores by environment
Aggbox<-ggplot(Agg, aes(x=Treatment, y= Behavioral_Response_Score, fill=Environment))+
  geom_boxplot(lwd=.6, width=.5, fatten=0.75,outlier.shape=NA, alpha=1, position=position_dodge(width=.7))+
  geom_point(aes(shape=Environment),position=position_jitterdodge(jitter.width=0.1, jitter.height=0.13))+
  #geom_jitter(aes(shape=Environment),position=position_dodge(0.6))+
```

```
  scale_fill_manual(values = c("blue3","cornflowerblue","paleturquoise1"), guide="none")+
  scale_shape_manual(values= c(16,17,15))+
  theme(panel.background = element_rect(fill = "white", colour = "black"))+
  theme(panel.border = element_rect(colour = "black", fill=NA, linewidth=2))+
  theme(panel.grid.major = element_line(colour = "grey100"))+
  theme(panel.grid.minor = element_line(colour = "grey100"))+
  scale_y_continuous(breaks = seq(0, 5, by = 1), limits = c(-0.3,5.3))+
  theme(panel.background = element_rect(fill = "white", colour = "black", size=1))+
  theme(panel.border = element_rect(colour = "black", fill=NA, size=1))+
  theme(axis.text = element_text(size=20,colour = "black"))+
  theme(axis.text.x = element_text(size=20, vjust=0,colour = "black"))+
  theme(axis.title.y = element_text(vjust=2, size=20))+
  theme(axis.title.x = element_text(size=20))+
  theme(plot.margin = unit(c(1,1,1,1), "cm"))+
  theme(legend.position="right")+
  theme(legend.title = element_text(size=20))+
  theme(legend.text = element_text(size=20))+
  ylab(expression(paste("Behavioral Response Score")))+
  theme(axis.title.x = element_blank())
Aggbox
```

Code for identifying statistical outliers in metabolic scaling data based on IQR

Note, outliers were excluded from the uploaded dataset as described in the methods. However, the below code is provided for transparency for how they were calculated. The values included in Tmin and Tmax were obtained from the summary statistics and would differ between variables.

##identify outliers in **brain metabolic rate**
summary(MR$**BrainuW**)
IQR(MR $**BrainuW**)
Tmin<- 0.7037 -(1.5*0.2520947)
Tmax<- 0.9558 +(1.5*0.2520947)
MR$**BrainuW**[which(MR$**BrainuW** > Tmax)]
MR$**BrainuW**[which(MR$**BrainuW** < Tmin)]

Code for centering fixed effects for general linear mixed models
Note, this was only used if the associated variable was a fixed effect in the model. For example, body mass was centered when it was used as a fixed effect in the model in which brain mass was the response variable, but body mass was not centered when body mass itself was used as a response variable.

Dataframe: worker metabolic scaling data
Abbreviation: MR

MR$Body_Metabolic_Rate_microWatts<-scale(MR$ Metabolic_Rate_microWatts, center = TRUE, scale = FALSE)
MR$body_mass_milligrams <-scale(MR$ body_mass_milligrams, center = TRUE, scale = FALSE)
MR$brain_mass_milligrams<-scale(M$brain_mass_milligrams, center = TRUE, scale = FALSE)
contrasts(MR$Treatment) <- contr.sum(3)
MR$Environment<-as.factor(M$Environment)
contrasts(MR$Environment) <- contr.sum(3)
Worker movement analysis

Dataframe: worker metabolic scaling data
Abbreviation: MR

##Identify movement averages by **treatment**
```
Movmeans<-MR%>%
  group_by(Treatment) %>%
  summarize(Mean = mean(Movement_millimeters), SD = sd(Movement_millimeters, na.rm=TRUE))
Movmeans
```

##GLMM with a negative-binomial distribution and treatment by environment interaction and non-interaction effects, with colony as a random effect and including AIC calculation
```
movmodel1<-glmmTMB(Movement_millimeters ~ Treatment*Environment + (1|Colony), data = MR,
                   zi = ~1,
                   family = nbinom2(link="log"))
movmodel2<-glmmTMB(Movement_millimeters ~ Treatment+Environment + (1|Colony), data = MR,
                     zi = ~1,
                     family = nbinom2(link="log"))
AIC(movmodel1,movmodel2)
anova<-Anova(movmodel1, type='III')
anova
summary(movmodel1)
```

##movmodel1 diagnostics
```
testUniformity(movmodel1)
testDispersion(movmodel1)
```

##Boxplot of worker movement organized by treatment and environment

```
Movbox<-ggplot(MR, aes(x=Treatment, y= Movement_millimeters, fill=Environment))+
  geom_boxplot(lwd=.6, width=.5, fatten=0.75,outlier.shape=NA, alpha=1, position=position_dodge(width=.7))+
  geom_point(aes(shape=Environment),position=position_jitterdodge(jitter.width=0.1, jitter.height=0))+
  #geom_jitter(aes(shape=Environment),position=position_dodge(0.6))+
  scale_fill_manual(values = c("blue3","cornflowerblue","paleturquoise1"), guide="none")+
  scale_shape_manual(values= c(16,17,15))+
  theme(panel.background = element_rect(fill = "white", colour = "black"))+
  theme(panel.border = element_rect(colour = "black", fill=NA, linewidth=2))+
  theme(panel.grid.major = element_line(colour = "grey100"))+
  theme(panel.grid.minor = element_line(colour = "grey100"))+
  scale_y_continuous(breaks = seq(0, 400, by = 50), limits = c(-1,500), labels = function(y) sub("1+$","1", y))+
  theme(axis.text = element_text(size=20,colour = "black"))+
  theme(axis.text.x = element_text(size=20, vjust=-1,colour = "black"))+
  theme(axis.title.y = element_text(vjust=6, size=20))+
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))+
  theme(legend.title = element_text(size=20))+
  theme(legend.text = element_text(size=20))+
  ylab(expression(paste("Distance (mm)")))+
  theme(axis.title.x = element_blank())+
  theme(legend.key.size = unit(1.5,'cm'))+
  guides(fill=guide_legend(title = "Environment", shape="Environment"))
Movbox
```

Statistical analyses of brain and body metabolic scaling variables including mass, metabolic rate, mass-specific metabolic rate, relative brain mass and relative brain metabolic rate

Note, the following R code provides an example of these statistical analyses performed on brain mass. Simply replace the bolded **relative brain metabolic rate** term with alternative variables as required for the analysis

```
##Calculate means and standard deviations for relative_brain_metabolic_rate for each treatment
relmeans<-MR%>%
  group_by(Treatment) %>%
  summarize(Mean = mean(relative_brain_metabolic_rate), SD = sd(relative_brain_metabolic_rate, na.rm=TRUE))
relmeans

##Calculate medians and IQR for relative_brain_metabolic_rate for each treatment
tapply(MR$relative_brain_metabolic_rate,MR$Treatment, summary)

##Constuct multiple general linear mixed models and dropping non significant terms, including models with a gaussian distribution, inverse-gaussian distribution, gaussian distribution of logi-transformed data, gaussian distribution of log transformed data, and gaussian distribution of square-root transformed data.

gaussian<-lmer(relative_brain_metabolic_rate ~ Treatment+
        Environment+body_mass_milligrams+ brain_mass_milligrams+Treatment*Environment+Treatment* body_mass_milligrams +
        (1|Colony), data = MR)
invgaus<-glmer(relative_brain_metabolic_rate ~ body_mass_milligrams+
        (Treatment*Environment)+(Treatment* body_mass_milligrams)+(1|Colony), data = MR, family=inverse.gaussian(link =
        "identity"))
logit<-lmer(logit(relative_brain_metabolic_rate, percents=FALSE,adjust=TRUE) ~Treatment+
        Environment+ body_mass_milligrams + brain_mass_milligrams +Treatment*Environment+Treatment* body_mass_milligrams
        +(1|Colony), data = MR)
```

```
log<-lmer(log(relative_brain_metabolic_rate)~Treatment+
        Environment+ body_mass_milligrams +brain_mass_milligrams+Treatment*Environment+Treatment* body_mass_milligrams
        +(1|Colony), data = MR)
sqrt<-lmer(sqrt(relative_brain_metabolic_rate)~Treatment+
        Environment+ body_mass_milligrams + brain_mass_milligrams +Treatment*Environment+Treatment* body_mass_milligrams
        +(1|Colony), data = MR)

##gaussian model diagnostics
testUniformity(gaussian)
testDispersion(gaussian)
testCategorical(gaussian,MR $Environment)
testCategorical(gaussian,MR $Treatment)
shapiro.test(resid(gaussian))

##ANOVA and eta-squared effect size calculation for the gaussian model
anova<-Anova(gaussian, type='III', test.statistic='F')
anova
eta_squared(anova)

##AIC calculation for model comparisons
AIC(gaussian, logit, log, sqrt)

##Pairwise comparisons for the gaussian model
brpairs<-emmeans(gaussian,"Treatment")
pairs(brpairs)
emmeans(sqrtbr1,pairwise~Treatment|Environment)
```

```
##Box plot for relative brain metabolic rate organized by treatment and environment
relmrbox<-ggplot(MR, aes(x=Treatment, y= relative_brain_metabolic_rate, fill=Environment))+
  geom_boxplot(lwd=.6, width=.5, fatten=0.75,outlier.shape=NA, alpha=1, position=position_dodge(width=.7))+
  geom_point(aes(shape=Environment),position=position_jitterdodge(jitter.width=0.1, jitter.height=0))+
  scale_fill_manual(values = c("blue3","cornflowerblue","paleturquoise1"), guide="none")+
  scale_shape_manual(values= c(16,17,15))+
  theme(panel.background = element_rect(fill = "white", colour = "black"))+
  theme(panel.border = element_rect(colour = "black", fill=NA, linewidth=2))+
  theme(panel.grid.major = element_line(colour = "grey100"))+
  theme(panel.grid.minor = element_line(colour = "grey100"))+
  theme(axis.text = element_text(size=20,colour = "black"))+
  theme(axis.text.x = element_text(size=20, vjust=-1,colour = "black"))+
  theme(axis.title.y = element_text(vjust=6, size=20))+
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))+
  theme(legend.title = element_text(size=13))+
  theme(legend.text = element_text(size=12))+
  theme(legend.key.size=unit(1.5,'cm'))+
   scale_y_continuous(labels = scales::percent,breaks = seq(0, 1, by =.2), limits = c(0,1.15))+
 ylab(expression(paste("Relative Brain Metabolic Rate")))+
  theme(axis.title.x = element_blank())+
  guides(fill=guide_legend(title = "Environment"), shape="none")
relmrbox

##Barplot of relative brain metabolic rate organized by Treatment
relbar<-ggbarplot(MR, x="Treatment", y="relative_brain_metabolic_rate", fill="white",
                  add = c("mean_sd"),
                  size=0.8, width = .5, add.params = list(size = 1))+
```

```
  geom_jitter(width=.15, aes(shape=Environment, fill=Environment), size=2)+
  scale_shape_manual(values= c(21,24,22))+
  scale_fill_manual(values = c("blue3","cornflowerblue","paleturquoise1"), guide="none")+
  scale_y_continuous(expand = c(0,0),labels = scales::percent,breaks = seq(0, 2, by =.2), limits = c(0,1))+
   #theme(panel.grid.major = element_line(colour = "grey80"))+
  #theme(panel.grid.minor = element_line(colour = "grey100"))+
  theme(panel.grid.major = element_blank())+
  theme(panel.grid.minor = element_blank())+
  theme(panel.background = element_rect(fill = "white", colour = "black", size=1))+
  theme(axis.text = element_text(size=20,colour = "black"))+
  theme(axis.text.x = element_text(size=20, vjust=0,colour = "black"))+
  theme(axis.title.y = element_text(vjust=2, size=20))+
  theme(axis.title.x = element_text(size=20))+
  theme(plot.margin = unit(c(1,1,1,1), "cm"))+
  theme(legend.position="none")+
  theme(legend.title = element_text(size=20))+
  theme(legend.text = element_text(size=20))+
  ylab(expression(paste("Relative Brain Metabolic Rate")))+
  theme(axis.title.x = element_blank())
relbar
```

##Standard Major Axis regression of **log brain metabolic rate** to **log brain mass** with pairwise comparisons of slope (note * before the Treatment term)

```
summary(sma(log(Brain_Metabolic_Rate_microWatts)~log(brain_mass_milligrams)*Treatment, MR, multcomp=TRUE,
        multcompmethod=c("adjusted")))
```

##Standard Major Axis regression of **log brain metabolic** rate to **log brain mass** with pairwise comparisons of y-intercept (note + before the Treatment term)

```
summary(sma(log(Brain_Metabolic_Rate_microWatts)~log(brain_mass_milligrams)+Treatment, MR, multcomp=TRUE,
        multcompmethod=c("adjusted")))
```

scatter plot of **log brain metabolic rate** and **log brain mass**. Note the ignored geom_abline() functions that were only included for significant scaling relationships, in which case slopes and intercepts are manually specified.

```
scatter<-ggplot(data=MRC_nooutlier,aes(x=log(brain.mass.mg), y=log(BrainuW), shape=Treatment))+
  geom_point(data=MRC_nooutlier, size=1.5)+
  scale_shape_manual(values= c(16,17,15))+
  #scale_fill_manual(values = c("cyan3","black"))+
  theme(panel.background = element_rect(fill = "white"),
        panel.grid.major = element_line(colour = "grey80"),
        panel.grid.minor = element_line(colour = "grey100"),
        panel.border = element_rect(colour = "black", fill=NA, linewidth=1),
        axis.text = element_text(size=15, colour = "black"),
        legend.text = element_text(size=11,margin = margin(t = 10, b = 10, unit = "pt")),
        legend.title = element_blank(),
        legend.position = "top",
        axis.title.x = element_text(vjust=-3, size=15),
        axis.title.y = element_text(vjust=5, size=15),
        plot.margin = unit(c(1, 1, 1, 1), "cm"))+
  scale_y_continuous(breaks = seq(-1.5, 0.5, by = .5), limits = c(-1.5,.5))+
  scale_x_continuous(breaks = seq(-4, -2.5, by =0.5), limits = c(-4,-2.5))+
  #geom_abline(intercept= ,slope= ,linewidth=, linetype=1)+
  #geom_abline(intercept= ,slope= ,linewidth=, linetype=2)+
  #geom_abline(intercept= ,slope= ,linewidth=, linetype=3)+
```

```
  ylab("log Brain Metabolic Rate (µW)")+
  xlab("log Brain Mass (mg)")
 #  guides(fill=guide_legend(title = "none"), linetype="Treatment")
 scatter
```

Scaling of log brain oxygen consumption rate to log brain mass across mammals and *Tetramorium immigrans*.

Note, importantly, the slope and confidence intervals of this scaling were calculated and placed only mammals, *after which* the point for *T. immigrans* was plotted. The slope and confidence intervals *were not* calculated including *T. immigrans*.

Dataframe: Mammal_scaling_data_figure_7
Abbreviation: Brainscaling

```
Mammals<-subset(Brainscaling,Class %in% c("Mammals"))
smaslope<-sma(log(brain_oxygen_consumption_rate_milliliters_per_min)~log(brain_mass_grams),Mammals)
summary(smaslope)
Scatter2<-ggplot(data=Mammals,aes(x=log(brain_mass_grams), y=log(brain_oxygen_consumption_rate_milliliters_per_min)))+
  geom_point(data=Mammals, size=.7)+
  scale_x_continuous(breaks = seq(-10, 8, by =2), limits = c(-11,8))+
  scale_y_continuous(breaks = seq(-13, 5, by =2), limits = c(-14,5))+
  stat_ma_line(method = "SMA",fullrange = TRUE, alpha=0.7)+
  theme(panel.background = element_rect(fill = "white", colour = "black"),
        panel.grid.major = element_line(colour = "grey90"),
        panel.grid.minor = element_line(colour = "grey100"),
        panel.border = element_rect(colour = "black", fill=NA, linewidth=1.5),
        axis.text = element_text(size=15, colour = "black"),
        legend.text = element_text(size=11,margin = margin(t = 10, b = 10, unit = "pt")),
        legend.title = element_blank(),
```

```
      axis.title.x = element_text(vjust=-3, size=15),
      axis.title.y = element_text(vjust=5, size=15),
      plot.margin = unit(c(1, 1, 1, 1), "cm"))+
  ylab(expression(paste("log Brain Oxygen Consumption Rate (ml/min)")))+
  xlab(expression(paste("log Brain Mass (g)")))+
  geom_point(data=Brainscaling, size=.7)
Scatter2
```